Vedant Desai

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EDUCATION

University of Michigan — Expected May 2028
B.S. in Computer Science and Economics — Ann Arbor, MI

- **GPA:** 3.66 / 4.0
- **Coursework:** Data Structures & Algorithms, Statistics and Data Analysis, Discrete Mathematics, Calculus III

EXPERIENCE

CaseStudyPrep.AI — Dec 2025 – May 2026
Software Engineer Co-op (Voice AI) — Remote

- Eliminated the silent-audio problem in a voice-AI product — most frames sent to Whisper were dead air — by running Silero VAD client-side via ONNX Runtime, filtering silence before upload and cutting cloud inference costs by 40%.
- Moved audio processing off the UI thread into a Web Worker with an async stream handoff, reducing main-thread blocking time to under 5ms and keeping the real-time audio visualizer at a smooth 60 FPS during active inference.

PROJECTS

SignalWeaver | *Python, PyTorch, LangGraph*
Multi-signal financial research platform with fine-tuned LLM sentiment and agentic scoring

- Lifted financial-sentiment classification accuracy from 81% to 96% by LoRA fine-tuning a quantized meta-llama/Meta-Llama-3.1-8B-Instruct model on 3,454 Financial PhraseBank entries, evaluated on a held-out test set.
- Orchestrated a 5-node LangGraph pipeline (fetch → classify → embed → score → explain) with per-node timing from the same 90-run batch: fetch 7.3s p50, classify 1.1s p50 (29ms/article over 40-article batches), embed 1.1s p50, score under 1ms p50, explain 3ms p50 — fetch dominated wall time (80% of p50 latency).

Vylet | *Python, LangGraph, Redis, Celery, Docker* — vyletdata.com
Automated lead-sourcing platform; live product generating $1,500 MRR across three paying clients

- Launched Vylet, automating a 30-minute manual process per business into a Dockerized LangGraph pipeline generating 30 scored leads in 30 minutes — a 30x speedup — with Redis/Celery workers on a recurring cycle.
- Engineered a LangSmith eval pipeline spanning 20 adversarial business test cases across 13 archetype labels — manufacturers, SaaS tools, PE holding companies, geographic mismatches — then layered deterministic Pydantic consensus gates to lift extraction faithfulness from 50% to 90%.

Granular Synthesizer Plugin | *C++, JUCE, SIMD, VST3* — github.com/Verdent06/granular-synth
Real-time audio DSP plugin in C++/JUCE with lock-free threading and VST3/AU binaries

- Implemented 16-voice polyphony with per-voice ADSR envelopes, pre-computing attack/decay/release rates as samples-per-step in prepareToPlay() to eliminate division inside the audio hot loop — and a voice-stealing algorithm that evicts the oldest active note by age counter when all 16 slots are busy, with each voice falling back to a sine oscillator until a WAV sample is loaded.
- Implemented a global post-mix delay on a power-of-two ring buffer (2^17 = 131,072 samples, O(1) bitmask addressing) with per-sample delay time smoothing — the read head steps ±1 sample per block toward the target delay value — eliminating the zipper noise audible when a user adjusts the delay knob during live playback.

Michigan Data Consulting (MDC) | *Python, Pandas, Flask, AWS EC2*
Campaign-finance data pipeline replacing manual PAC research for a nonprofit stakeholder

- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at 2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating 800 hours of manual pulls across 400 tracked PACs.
- Architected a deterministic aggregation engine that parses those exports, normalizes irregular contribution amounts, and ranks PACs by total funding volume so researchers stop rebuilding spreadsheets to surface top spenders.

TECHNICAL SKILLS

Languages .... Python, C++
AI / ML .... PyTorch, LangGraph, LangSmith, ONNX Runtime
Databases .... Redis
Cloud & Infrastructure .... Docker, Celery, Git, AWS EC2